SMART MODULATION ADVISOR

Website Interface & Feature Blueprint

A design plan for an interactive digital modulation advisor

PROJECT IDEA
An interactive application that uses the selected Signal-to-Noise Ratio (SNR) and a Bit Error Rate (BER) requirement to recommend the highest-data-rate modulation scheme that meets the reliability target.

1. Design Direction

The Smart Modulation Advisor should look like a professional wireless-communications simulator rather than a basic webpage. Its interface should be modern, clear, interactive, and suitable for a college project demonstration.

Element	Recommended design
Background	Dark navy
Primary accent	Electric blue
Secondary accent	Purple
Highlight color	Pink
Successful recommendation	Green
Cards	Rounded corners with subtle borders
Graphs	Clear, contrasting lines and labelled axes
Typography	Modern, readable sans-serif
Overall feel	Scientific, futuristic, professional

2. Sidebar Navigation

Keep the sidebar consistent across every page. Use a narrow navigation rail with an icon and label for each section, and highlight the active page with a purple background or subtle blue glow.

Sidebar item	Contents	Priority
Home	Main dashboard, current SNR, BER graph, recommendation	Essential
Simulator	Interactive controls and modulation experiments	Essential
Performance Analysis	BER curves, data rate, reliability comparison	Essential
Modulation Theory	BPSK, QPSK, 16-QAM explanations and constellation diagrams	Essential
Insights	Adaptive switching points and performance observations	Recommended
Settings	BER threshold, graph preferences, simulation parameters	Essential
About Project	Overview, objectives, methodology, technologies	Recommended

3. Page-by-Page Feature Plan

Page 1 — Home Dashboard

Purpose: answer immediately what the current signal quality is, which modulation scheme is recommended, and why.

• Header: app logo, project name and tagline, simulator status, optional light/dark toggle.

• SNR control: slider to change signal quality.

• BER vs SNR graph: curves for BPSK, QPSK and 16-QAM.

• Recommendation card: selected scheme, calculated BER, data rate and explanation.

• Data-rate indicator: bits per symbol for the selected scheme.

• Adaptive switching graph: shows how the recommended scheme changes with SNR.

• Quick comparison cards: concise strengths and limitations of each scheme.

Example recommendation: “QPSK is recommended at the current SNR because it provides the highest data rate among the schemes that satisfy the selected BER limit.”

Page 2 — Simulator

Purpose: let users change parameters and run experiments.

Feature	Purpose
SNR slider	Change signal quality
BER threshold	Set maximum acceptable error rate
Modulation selector	Choose BPSK, QPSK, 16-QAM or compare all
Number of bits	Control the size of a Monte Carlo simulation
Run Simulation	Start calculation
Reset	Restore default parameters
Results panel	Show calculated BER values

Optional quick presets: Low (4 dB), Medium (10 dB), and High (18 dB). These are demonstration presets, not guaranteed switching thresholds; the actual recommendation depends on BER calculations.

Page 3 — Performance Analysis

• BER comparison graph with SNR (dB) on the x-axis and BER on the y-axis.

• Separate curves for BPSK, QPSK and 16-QAM.

• Horizontal line showing the selected BER limit.

• Performance table covering bits per symbol, spectral-efficiency assumptions, noise tolerance and data-rate potential.

• Additional metrics where implemented: SER, estimated throughput, and whether each scheme meets the selected threshold.

Parameter	BPSK	QPSK	16-QAM
Bits per symbol	1	2	4
Ideal spectral efficiency*	1 bit/s/Hz	2 bits/s/Hz	4 bits/s/Hz
Data-rate potential	Low	Medium	High
Reliability at comparable conditions	Generally robust	Intermediate	Needs better signal quality

*Ideal spectral-efficiency values assume ideal symbol-rate and bandwidth conditions, without coding or overhead.

Page 4 — Modulation Theory

BPSK — Binary Phase Shift Keying: 1 bit per symbol; two possible signal phases; suitable for robust communication.

QPSK — Quadrature Phase Shift Keying: 2 bits per symbol; four distinct signal states; balances data rate and reliability.

16-QAM — 16-state Quadrature Amplitude Modulation: 4 bits per symbol; 16 constellation points; higher spectral efficiency but generally requires better signal quality for comparable error performance.

Visual feature: include constellation diagrams for BPSK, QPSK and 16-QAM to show how their signal points differ.

Page 5 — Insights

• Adaptive switching graph: a step graph showing the recommended scheme as SNR increases.

• Switching-point calculator: display the SNR at which each scheme becomes eligible under the selected BER limit.

• Recommendation explanation: explain why a scheme was chosen and why faster alternatives may fail the requirement.

• Signal-quality label: Poor, Moderate, Good or Excellent, based on clearly defined ranges.

• Simulation summary: show how often each scheme meets the error requirement across the tested SNR range.

Signal-quality labels are illustrative categories, not universal standards.

Page 6 — Settings

• Default BER threshold.

• Minimum and maximum SNR.

• Number of bits per simulation.

• Simulation method: theoretical, Monte Carlo, or both, if implemented.

• Graph display preferences.

• Reset to default settings.

For the first version, user accounts, cloud storage and a database are unnecessary. Browser-local settings are sufficient.

Page 7 — About Project

• Project overview and problem statement.

• Objectives and working principle.

• Technologies used.

• Assumptions and limitations.

• Future scope.

• Team members, if applicable.

Explain how adaptive modulation relates to wireless communications, but do not claim that the prototype directly controls a real Wi-Fi, 4G or 5G connection. It demonstrates the principle through simulation.

4. Additional Visual Mockups to Plan

A. Simulator Workspace: Input controls, modulation selection, simulation progress and numerical results.

B. Modulation Comparison: Constellation diagrams, BER curves, a performance table, and a reliability-versus-spectral-efficiency comparison.

C. Insights and Switching: Adaptive switching graph, calculated transition points, recommendation explanation and simulation summary.

Together with the Home Dashboard, these screens provide a strong first interface plan.

5. Development Roadmap

Phase	Features
Phase 1 — Core project	SNR slider, BER calculations, modulation recommendation, BER graph and data rate
Phase 2 — Strong demonstration	Modulation comparison, adjustable BER threshold, switching-point graph and constellation diagrams
Phase 3 — Advanced features	Theoretical vs simulated BER, downloadable results, noise-model selection and additional modulation schemes

6. Recommended Technology Stack

• HTML — page structure.

• CSS — layout, dark theme, cards and responsive design.

• JavaScript — SNR controls, recommendation logic and interactive behavior.

• Chart.js — BER curves and performance graphs.

• Python (optional) — reuse an existing Python simulator through a backend, if appropriate.

If an existing simulator is written in Python, inspect its inputs and outputs before deciding whether to connect it through a backend or implement equivalent calculations in JavaScript.

7. Technical Rules for a Correct Advisor

• Recommend the highest-data-rate modulation scheme whose calculated BER is at or below the selected BER threshold.

• Recalculate the recommendation whenever SNR or the BER threshold changes.

• Do not use fixed switching points unless validated for the chosen BER model, SNR definition and assumptions.

• Clearly label theoretical BER results versus Monte Carlo simulation results.

• Show the calculated BER and explain why the selected scheme qualifies.

• Keep graph axes, units, legends and thresholds clearly labelled.

8. Final Sidebar Layout

■ SMART MODULATION ADVISOR

■ Home
■ Simulator
■ Performance Analysis
■ Modulation Theory
■ Insights
■■ Settings
■■ About Project

Adaptive. Reliable. Efficient.

Final Design Recommendation

Use a consistent dark navy background, blue-purple accents, rounded cards, readable typography and clear charts throughout. Keep the recommendation card visually prominent. Build the core simulation first, then add the more advanced analysis screens.

The project’s central rule is simple: choose the fastest modulation scheme that satisfies the user’s BER requirement at the current SNR. The calculations—not arbitrary visual thresholds—must drive the recommendation.